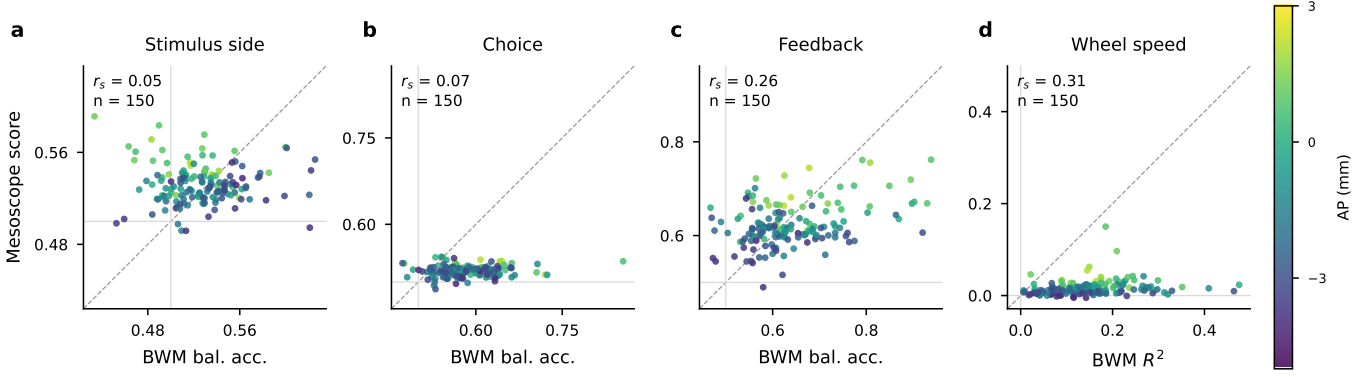


**Figure 1:** Decoding per spatial bin, BWM Neuropixels (x) vs mesoscope (y), decoded from 10 randomly drawn neurons of the session in the bin (mean of 3 draws). Each point is one hexagonal bin (0.4 mm, hemispheres folded) with at least one session of each modality contributing $\geq$10 neurons; its value is the score averaged across those sessions (same values as the decoding maps). Colour: AP position of the bin. Dashed line: identity; grey lines: chance (0.5 balanced accuracy, 0 for $R^2$). The $p$-values ignore spatial autocorrelation between neighbouring bins and are therefore optimistic. **a** Stimulus side (Spearman $r_s = 0.05$, $p = 0.55$, 150 bins). **b** Choice (Spearman $r_s = 0.07$, $p = 0.37$, 150 bins). **c** Feedback (Spearman $r_s = 0.26$, $p = 0.0013$, 150 bins). **d** Wheel speed, cross-validated $R^2$ (Spearman $r_s = 0.31$, $p = 9.1 \times 10^{-5}$, 150 bins).